

Auteur: Abdellah El Moussaoui

Studierichting: Informatica

Oefeningenreeks 5G nr. 1.13

$$V = \pi \int_0^1 (\sqrt{x} \sqrt[4]{1-x})^2 dx$$

$$V = \pi \int_0^1 x (1-x)^{\frac{1}{2}} dx$$

$$\text{Laat } t = (1-x)^{\frac{1}{2}}$$

$$x = 1 - t^2 \Rightarrow dx = -2t dt$$

$$V = -2\pi \int_1^0 (1-t^2) t^2 dt$$

$$= 2\pi \int_0^1 t^2 - t^4 dt$$

$$= 2\pi \left[\frac{1}{3} t^3 - \frac{1}{5} t^5 \right]_0^1$$

$$= \frac{4}{15} \pi$$

References

5G-1.13-abdellah.el.moussaoui.INF.png